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Fourth-generation Drones

DOI: 10.1201/9788743812418-5

5.1 Introduction

The fourth generation of drones represents a defining leap in the evolution of unmanned aerial systems, marking the transition from semi-autonomous platforms to intelligent, fully autonomous, and network-integrated aerial systems. This generation builds upon the endurance, stealth, and multi-role capabilities of third-generation UAVs, integrating artificial intelligence (AI), machine learning (ML), advanced swarm coordination, and real-time data analytics to enable self-governing decision-making and collaborative mission execution. Unlike their predecessors, fourth-generation drones are not merely remotely piloted or pre-programed systems; they are adaptive, cognitive platforms capable of learning, optimizing, and responding dynamically to changing mission environments without constant human intervention.

Technological advancements in AI-driven autonomy, edge computing, 5G-enabled communication, and quantum-encrypted data links have redefined how drones operate within both military and civilian domains. These systems exhibit multi-domain interoperability, seamlessly coordinating across air, land, sea, and space-based assets within network-centric warfare frameworks. Militarily, fourth-generation drones perform swarm intelligence operations, loyal wingman missions, and AI-assisted